AI Adoption for RTI, Tutoring, and Student Outcomes

A Research Synthesis for Education Administrators

Prepared: February 14, 2026

Greeneville City Schools

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1. Definition & Background

Key Terms

Response to Intervention / RTI (Tennessee: RTI²)

A multi-tiered framework for identifying and supporting students with learning needs through increasingly intensive levels of instruction and intervention. Tennessee's RTI² framework uses three tiers: Tier I (high-quality core instruction for all), Tier II (targeted learning acceleration for some), and Tier III (intensive skills-based intervention for a few). Universal screening of all K-8 students occurs in reading, mathematics, and written expression.

Multi-Tiered System of Supports (MTSS)

A broader framework that encompasses RTI and adds behavioral and social-emotional supports. MTSS integrates academic intervention (RTI) with Positive Behavioral Interventions and Supports (PBIS) and social-emotional learning (SEL) into a unified system.

AI Tutoring

Computer-based instructional systems that use artificial intelligence to provide personalized, one-on-one or small-group academic support. This includes two generations of technology: (1) Intelligent Tutoring Systems (ITS), which use rule-based or machine learning algorithms to adapt instruction based on student responses (e.g., Carnegie Learning, DreamBox, IXL), and (2) Generative AI tutors, which use large language models to provide conversational, Socratic-style tutoring (e.g., Khanmigo, custom GPT-based tutors).

High-Dosage Tutoring (HDT)

Tutoring delivered at least three times per week in small groups (typically 1-3 students per tutor), during the school day, by a consistent tutor. This model has the strongest evidence base in education research. Tennessee's TN ALL Corps program followed this model, serving over 225,000 students between 2021 and 2024.

Historical Context

The aspiration to use technology to replicate one-on-one tutoring dates to Benjamin Bloom's landmark 1984 paper, "The 2 Sigma Problem." Bloom found that students receiving individual tutoring with mastery learning performed 2.0 standard deviations above conventionally-taught peers—placing the average tutored student above 98% of the control group. The "problem" was that human tutoring was too expensive to scale.

Bloom, B. S. (1984). "The 2 Sigma Problem." Educational Researcher, 13(6), 4-16.

More recent re-analysis by VanLehn (2011) suggests Bloom's 2.0 effect was inflated. The actual effect of human tutoring is closer to d = 0.79, and step-based intelligent tutoring systems achieve d = 0.76—nearly equivalent. This was a landmark finding: computer tutors were closer to human tutors than anyone assumed.

VanLehn, K. (2011). "The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems." Educational Psychologist, 46(4), 197-221.

The release of ChatGPT in November 2022 launched a new wave of AI tutoring. Generative AI tutors can engage in natural conversation, explain concepts in multiple ways, and adapt in real time—capabilities that traditional ITS could not achieve. By 2025, platforms like Khanmigo (Khan Academy), Tutor CoPilot (Stanford), and LearnLM (Google DeepMind) had entered randomized controlled trials, and over 700,000 students were using AI tutoring tools in the U.S.

2. Evidence of Effectiveness

Meta-Analyses of Intelligent Tutoring Systems

Three major meta-analyses establish the baseline for ITS effectiveness:

Meta-Analysis	Studies/N	Overall Effect	Key Finding
Ma et al. (2014)	107 effects / 14,321 students	g = 0.43	ITS outperform teacher-led instruction and other computer-based instruction, but NOT human tutoring
Kulik & Fletcher (2016)	50 evaluations	d = 0.66 (median)	Raises scores from 50th to 75th percentile; larger effects on curriculum-aligned tests
VanLehn (2011)	Multiple experiments	d = 0.76 (ITS) vs. 0.79 (human tutoring)	Step-based ITS nearly as effective as human tutoring

Ma, W. et al. (2014). Journal of Educational Psychology, 106(4), 901-918.

Kulik, J. & Fletcher, J. (2016). Review of Educational Research, 86(1), 42-78.

Recent Meta-Analyses (2024-2025)

The generative AI era has produced its own meta-analyses, with larger but more variable effect sizes:

- Overall AI in Education (2025, Computers and Education Open): d = 0.924 across 29 studies and 2,657 participants. However, heterogeneity was very high (I^2 = 93%), meaning outcomes vary enormously by context.
- Generative AI and Learning Outcomes (2025): SMD = 0.45 across 68 experimental studies. Chatbot-based interventions showed the largest gains (ES = 1.02).
- AI vs. Human Feedback (2025, Educational Psychology): No statistically significant differences across 41 studies and 4,813 students.
- K-12 ITS Review (2025, npj Science of Learning): Effects "generally positive but mitigated when compared to non-intelligent tutoring systems" across 28 studies and 4,597 K-12 students.

Major Randomized Controlled Trials

Large-Scale K-12 RCTs (Established Platforms)

Platform	Year	N	Grades	Subject	Effect Size	Notes
Carnegie Learning	2014	18,743	HS (Algebra)	Math	d = 0.21 (Year 2)	No Year 1 effect; lowest quartile gained 11 percentile points
ASSISTments (Maine)	2016	Large	Grade 7	Math	d = 0.18	Less than $100/student;

						effects persist 1 year post-program
ASSISTments (NC replication)	2024	Large	Grade 8	Math	d = 0.10	31% improvement over expected annual gain
Khan Academy	2024	~11,000	Grades 3-8	Math	d = 0.12-0.22	RCT by University of Toronto; published in PNAS
IXL	Multiple	Large	Grades 3-8	Math	~d = 0.20	ESSA Tier 2 (quasi-experimental, not RCT)

Pane, J. et al. (2014). Educational Evaluation and Policy Analysis, 36(2), 127-144.

Roschelle, J. et al. (2016-2024). ASSISTments RCTs. assistments.org/evidence-of-impact

Oreopoulos, P. et al. (2024). PNAS. doi.org/10.1073/pnas.2507708123

Generative AI Tutoring RCTs (2024-2025)

Study	Year	N	Population	Subject	Effect Size	Key Finding
Kestin et al. (Harvard)	2025	194	College (physics)	Physics	d = 0.73-1.30	AI tutor outperformed active learning; 20% less time spent
Bastani et al. (Wharton/Penn)	2025	~1,000	HS (Turkey)	Math	GPT Tutor: +127%; GPT Base: -17%	Unguarded AI HARMED learning on independent tests
Wang et al. (Tutor CoPilot)	2024	1,800 students	K-12	Math	+4pp mastery (+9pp for weak tutors)	AI assists human tutors, not students directly; $20/tutor/yr
Google LearnLM/Eedi	2025	165	Secondary (UK)	Math	+5.5pp (OR=1.3)	Human-supervised AI; 0.1% hallucination rate
Schmucker et al. (Italy)	2024	4 classes	HS (Italy)	ESL/Grammar	Significant	GPT-4 homework tutor improved engagement and outcomes

Kestin, G. et al. (2025). Scientific Reports. nature.com/articles/s41598-025-97652-6

Bastani, H. et al. (2025). PNAS. doi.org/10.1073/pnas.2422633122

Wang, R. et al. (2024). EdWorkingPaper No. 24-1054. Stanford SCALE.
Google DeepMind & Eedi (2025). storage.googleapis.com/deepmind-media/LearnLM/learnLM_nov25.pdf

AI for Reading Intervention

Amira Learning (HMH)

Amira is an AI-powered reading tutor that uses speech recognition to listen to students read aloud, diagnose errors in real time, and provide micro-interventions. A large-scale study across 12 Louisiana districts found:

- Effect size of d = 0.40 overall on DIBELS—twice as effective as traditional tutoring
- 68% faster reading growth than other reading technologies
- Students with frequent use showed 2x weekly growth vs. non-users
- Reading growth comparable to high-dosage human tutoring at 1/70th the cost
- ESSA Level II evidence rating

Evidence for ESSA: evidenceforessa.org/program/amira/

Lexia Core5 / PowerUp (Cambium)

Lexia has the most established evidence base among adaptive reading platforms, with 20 peer-reviewed studies and a Strong ESSA rating. PowerUp (grades 6-12) was found to be up to 5x more effective than average reading intervention programs in a recent quasi-experimental study.

Lexia Learning: lexialearning.com/research/product-efficacy

3. What Works vs. What Doesn't

What Works

1. Pedagogical Design with Guardrails

The single most important finding across all studies: the same AI technology produces dramatically different outcomes depending on design. In the Bastani et al. study, standard GPT-4 ("GPT Base") caused a 17% decline in independent test performance, while the same GPT-4 with pedagogical guardrails ("GPT Tutor") produced a 127% improvement during practice with no harm on independent tests. Effective guardrails include Socratic questioning (asking guiding questions rather than giving answers), mastery checks before advancing, scaffolded hints, and metacognitive prompts.

2. Human-AI Hybrid Models

The strongest and most consistent evidence supports AI assisting human tutors rather than replacing them. Stanford's Tutor CoPilot ($20/tutor/year) showed that AI-assisted human tutors produced a 4 percentage point increase in math mastery overall, with a 9 percentage point increase for students of lower-rated tutors. Google's LearnLM study found human-supervised AI tutoring slightly outperformed human-only tutoring (66.2% vs. 60.7% on knowledge transfer) with a 0.1% hallucination rate. The greatest benefit goes to students with the least experienced tutors, suggesting AI can close the quality gap.

3. Sufficient Dosage

Across platforms, effects only appear with consistent usage. Khan Academy research showed gains at 30+ minutes per week. Carnegie Learning effects took a full year to materialize. Amira results were tied to "frequent use." Inconsistent or low-dosage implementation is the most common reason AI tools fail to produce results.

4. Integration into Existing RTI/MTSS Workflows

AI tools that integrate with existing student information systems and assessment platforms (Branching Minds, Panorama, PowerSchool) are more likely to be used effectively than standalone products. Tools that reduce team workload rather than adding another system to manage show higher adoption.

5. Purpose-Built MTSS Tools over Generic AI

For student-level decision-making (tier placement, intervention selection, progress monitoring), purpose-built MTSS platforms that draw on research-vetted intervention libraries are far more appropriate than general-purpose AI. As Branching Minds noted: "Generic AI pulls from internet noise, not research-vetted intervention strategies."

What Doesn't Work

1. Unguarded Access to Generative AI

Giving students unrestricted access to ChatGPT or similar tools for academic work creates dependency. The Bastani study showed students used GPT-4 as a "crutch" and performed 17% worse when it was removed. Multiple studies confirm a pattern of "cognitive offloading"—the less students rely on their own reasoning, the less capable they become.

2. AI Without MTSS Infrastructure

Purchasing AI tools before establishing basic MTSS processes (team structures, meeting protocols, data review cycles) means the AI generates data and recommendations that no one acts on. AI amplifies what you already have—if your foundation is weak, AI will not fix it.

3. One-Time Teacher Training

Teachers who receive a single training session on an AI platform rarely use it effectively. Only 29% of teachers have received training on effective AI tool use. Effective training requires ongoing support, addresses initial fear and discomfort before tool instruction, and uses professional learning communities for sustained implementation.

4. Expecting Lab-Scale Results at District Scale

This is critical: when samples are aligned with large-scale U.S. tutoring programs, pooled effect sizes shrink to one-third to one-half of those from controlled studies. Lab results of d = 0.76-1.3 should be expected to deliver d = 0.25-0.65 in real district implementation. Plan accordingly.

5. Using Generic AI for MTSS Decisions

Entering student data into ChatGPT or other non-compliant AI systems for tier placement or intervention planning risks FERPA violations and produces recommendations with no grounding in MTSS research or actual student data.

4. Criticisms & Limitations

Cognitive Risks

Cognitive Offloading and Dependency

A systematic review in Smart Learning Environments (Zhai et al., 2024) found that regular use of AI tutoring systems is associated with decline in critical thinking, decision-making, analytical reasoning, and information retention. An MIT Media Lab study (2025) using EEG monitoring found students using ChatGPT showed lower brain activity, weaker memory recall, and less ownership of their writing. Researchers coined the term "cognitive debt"—the accumulated cost of outsourcing thinking to AI. The meta-study in Cogent Education (2025) confirmed that greater AI dependence was associated with lower critical thinking levels, with cognitive fatigue partially mediating this relationship.

Zhai, C. et al. (2024). Smart Learning Environments, 11, Article 28.

MIT Media Lab (2025). "Your Brain on ChatGPT." media.mit.edu/publications/your-brain-on-chatgpt/

Equity and Access

The Digital Divide

A Frontiers in Computer Science (2026) article found that rural districts often contend with unreliable electricity and limited connectivity. Brookings (2024) warned about a "next digital divide" where the gap shifts from technology access to quality of AI use—affluent districts getting well-implemented AI tools while under-resourced districts get lower-quality versions.

Algorithmic Bias

Stanford's Center for Racial Justice (2024) documented that AI algorithms using historical data are more prone to discriminatory outcomes. A 2021 study found that 80% of AI systems in education showed some form of bias when not properly audited. In MTSS contexts, bias manifests as poor recall (missing at-risk students from disadvantaged groups) and poor precision (over-identifying students for unnecessary interventions).

Stanford Law School (2024). law.stanford.edu/2024/06/29/how-will-ai-impact-racial-disparities-in-education/

OECD Digital Education Outlook (2023). Algorithmic Bias chapter.

Teacher Training Equity

RAND found that low-poverty districts continue to outpace higher-poverty counterparts in training teachers on AI use. The students most likely to benefit from AI tutoring are least likely to have well-prepared teachers to implement it.

Data Privacy

In January 2025, the FTC finalized COPPA amendments shifting from opt-out to opt-in consent, with penalties up to $51,744 per affected child. About 31 states have published AI guidance for K-12 as of December 2025. Districts must verify FERPA compliance for all AI platforms and

ensure student data is not used to train external models. Tennessee's Public Chapter 550 requires all districts to have AI policies in place.

FTC COPPA 2025 Amendments. Future of Privacy Forum: fpf.org/wp-content/uploads/2024/10/Ed_AI_legal_compliance.pdf

Relationship and Engagement Concerns

EdWeek (2025) reported that AI risks hurting students' ability to develop meaningful relationships with teachers. AI expert John Bailey cautioned about "simulated relationships," noting that students may prefer the affirming relationships they can have with AI systems, creating a cycle where disconnection breeds more disconnection. Brookings emphasized there is little to no evidence that generative AI tutors can match human tutoring in emotional responsiveness, motivational support, and detecting frustration or anxiety.

EdWeek (2025). edweek.org/technology/rising-use-of-ai-in-schools-comes-with-big-downsides-for-students/2025/10

Long-Term Evidence Gap

This is the most significant limitation in the research: nearly all positive AI tutoring studies are short-term (days to weeks). No published long-term (semester or year-long) RCTs of AI tutoring in K-12 exist as of early 2026. The Harvard physics study was a single class session. The Google LearnLM study was 7 weeks. The possibility of novelty effects wearing off and cognitive dependency developing over time remains untested.

Students with Disabilities

57% of special education teachers reported using AI for IEPs during 2024-25. However, advocates have raised concerns about IDEA violations, privacy risks, and AI-generated IEPs undermining legally mandated individualized services. Districts cannot substitute AI tutoring for mandated one-on-one human support unless explicitly permitted within an IEP. A systematic review of AI for students with learning disabilities (2025) found all 11 included studies reported positive outcomes, but 70% had moderate risk of bias and 30% had high/serious risk.

EdWeek (2025). Teachers Are Using AI to Help Write IEPs.

Paglialunga et al. (2025). PMC/Brain Sciences.

5. Tennessee Context

State Legislation

Public Chapter 550 (HB 1630) — AI Policy Requirement

Signed into law spring 2024, Tennessee became one of the first states to require AI policies in schools. All local boards of education, public charter schools, and public universities must adopt policies regarding AI use by students, faculty, and staff for instructional and assignment purposes. Policies were required by July 1, 2024, for the 2024-25 school year. No additional state funding was attached.

publications.tnsosfiles.com/acts/113/pub/pc0550.pdf

HB 825 — Teen Social Media and Internet Safety Act (2025)

Signed by Governor Lee in April 2025. Directs TDOE to develop guidance for grades 6-12 on social media, internet, and AI safety. Guidance due by January 2026; implementation in the 2026-27 school year. LEAs must block social media on school internet unless allowed by a teacher for educational purposes.

wapp.capitol.tn.gov/apps/BillInfo/Default?BillNumber=HB0825&GA=114

2025 Companion Statute

Directs TDOE to provide guidance and fund professional development on responsible AI integration in schools.

Tennessee's RTI² Framework

Tennessee's RTI² uses three tiers with universal screening of all K-8 students. TDOE hosts monthly communities of practice focusing on screening, dyslexia characteristics, data-based decision making, and intervention supports. There is no formal TDOE guidance specifically addressing AI tools within RTI² as of February 2026. However, AI tools are being used by districts at all three tiers:

- Tier I: AI tools like MagicSchool help teachers create differentiated, standards-aligned lessons
- Tier II: AI platforms like Khanmigo and CourseMojo provide 15-20 minute embedded lessons with instant feedback
- Tier III: Sumner County's CourseMojo pilot showed special education students narrowing the achievement gap by two-thirds

TDOE RTI²: tn.gov/education/districts/instruction/tdoe-rti2.html

TISA Funding for Tutoring

The Tennessee Investment in Student Achievement (TISA) formula includes $376 million in direct funding for specific student needs. Rising 4th-grade students who score "below" on ELA TCAP receive $500 per student for tutoring and literacy support. Tennessee is the only state that

has integrated high-impact tutoring into its ongoing funding formula through the Learning Loss Remediation and Student Acceleration Act.

TISA Guide: tn.gov/content/dam/tn/education/tisa-resources/2025-26_TISA_Guide.pdf

TN ALL Corps and Post-ESSER Landscape

Between 2021 and 2024, Tennessee served more than 225,000 students through the TN ALL Corps high-dosage tutoring program. Districts and TDOE invested approximately $200 million in federal funds ($700/student from ESSER + $800 district match). ESSER III funds expired September 30, 2024 (with liquidation extensions to March 2026). The critical question now is sustaining tutoring programs post-ESSER—AI tools offer a cost-effective path.

Tennessee Districts Using AI

District	Size	Platform	Use Case	Results
Hamilton County	~45,000	Khanmigo	Math & ELA tutoring (Year 2.5)	Adoption described as "astounding"
Sumner County	~30,200	CourseMojo	6th-8th ELA intervention	8% TCAP increase (2x state avg); SpEd gap narrowed by 2/3
Collierville	~8,600	MagicSchool	Standards-aligned content	Student-teacher tech advisory vets tools
Sevier County	~15,000	Various	Administrative tasks	Reduced admin burden

Relevance to Greeneville City Schools (~2,500 students)

No Tennessee districts of comparable size have publicly documented AI tutoring implementations. Collierville (~8,600) is the smallest documented. However, GCS has strong positioning:

- TCAP scores outperform state averages in every content area
- Four-year gains: ELA +19 pts, Math +20 pts, Science +18 pts, Social Studies +26 pts
- The Niswonger Foundation, headquartered in Greeneville, serves 18 school systems in NE Tennessee and focuses on STEM/technology education
- GCS could be a model for small-district AI implementation in Tennessee

GCS TCAP Results: www4.gcschools.net/article/2303995

SCORE Survey Data (Spring 2025)

SCORE partnered with TOSS and the Tennessee State Board of Education to survey all 147 districts (86 respondents):

- 85% reported educators are using AI; nearly 25% use it "often"
- 84% see reducing administrative tasks as the biggest benefit
- 75% see AI supporting personalized learning

- 67% see improved student engagement; 63% see enhanced learning outcomes
- More than 60% of districts actively use AI tools for teacher workload, personalized learning, and operations

SCORE: tnscore.org/perspectives-and-press/perspectives/survey-findings-capture-ai-use-in-tennessee-school-districts

TEAM Evaluation and TVAAS

There is no published TDOE guidance on how AI tool use affects TVAAS calculations or TEAM evaluation scores. However, AI tools directly support several TEAM rubric indicators: differentiation of instruction, student engagement, checks for mastery, and data-driven instruction. If AI tutoring improves TCAP performance, this should be reflected in teacher and school TVAAS growth scores. Districts should position AI as supporting teacher-led instruction—ensuring the teacher remains the active instructional leader during observations.

TEAM rubric: team-tn.org/wp-content/uploads/2013/08/TEAM-General-Educator-Rubric.pdf

6. Measurable Outcomes

Test Scores and Achievement

The strongest and most consistent outcome linked to AI tutoring is improvement on standardized math assessments. Effect sizes range from d = 0.10 (ASSISTments replication) to d = 1.30 (Harvard physics, single session) in RCTs. At K-12 scale with thousands of students, realistic effect sizes are d = 0.10-0.22. In Tennessee, Sumner County's CourseMojo pilot produced an 8% year-over-year TCAP increase, more than double the statewide average.

Equity Gap Effects

Evidence is mixed but leans toward AI having greater benefit for lower-performing students:

- Tutor CoPilot: +9 percentage points for students of lower-rated tutors vs. +4 points overall
- Cross-sector research: below-average performers improved 43% with AI vs. 17% for above-average
- Khan Academy gains were consistent across demographic groups
- Sumner County: special education students narrowed the achievement gap by two-thirds

However, infrastructure gaps, algorithmic bias, and teacher training inequities mean AI could widen between-district gaps even while narrowing within-district gaps without deliberate equity-focused implementation.

Engagement and Time-on-Task

Students using AI tutors consistently report higher engagement and motivation. The Harvard study found students spent 20% less time on AI lessons while learning significantly more. However, researchers caution this may be a novelty effect—no study has confirmed sustained engagement beyond a few weeks.

Attendance

High-impact tutoring (not AI-specific) is associated with students being 1.2 percentage points less likely to be absent on tutoring days, with the strongest effects for small groups of 1-2 students at 3-5 sessions per week.

EdWorkingPapers (2024). edworkingpapers.com/policy-practice-series/ai24-1107

Long-Term and Graduation Outcomes

No published evidence exists linking AI tutoring specifically to graduation rates or long-term academic outcomes. This remains a major gap. The longest follow-up is the ASSISTments North Carolina replication, which showed effects persisting one year post-intervention (d = 0.10 on state math test at end of 8th grade, after 7th grade intervention).

Differential Effects by Subgroup

Subgroup	Finding	Evidence Strength

Low-achieving students	Benefit most from AI tutoring and human-AI hybrids	Moderate (RCT + quasi-experimental)
Students with disabilities	All 11 studies in a systematic review showed positive outcomes; IDEA compliance concerns	Limited (high bias risk in studies)
ELL students	Translation and adaptive language practice promising; standardized responses may not adapt	Limited (mostly ESL studies)
Low-income students	May benefit most, but implementation quality gap works against them	Theoretical + observational
Students of color	Could close gaps, but algorithmic bias risks reinforcing historical disparities	Mixed; bias well-documented

7. Implementation Models

Scheduling Models

Model	Description	Evidence
During School Day (Embedded)	15-20 min lessons integrated into class 3x/week	Strongest effects; Sumner County model; reduces absence
AI-Assisted Human Tutoring	AI provides real-time suggestions to human tutors	Tutor CoPilot RCT: +4pp mastery; $20/tutor/yr
Homework/At-Home Support	Students use AI tutor independently at home	Harvard study: strong results; engagement concerns for younger students
Intensive AI Model	2 hrs AI academics + enrichment afternoon	Alpha School claims top 1-2% nationally; NOT independently verified

The Human-AI Hybrid Model (Recommended)

The strongest evidence supports a model where AI assists human educators rather than replacing them:

- AI handles: adaptive content delivery, instant feedback on practice problems, progress monitoring data collection, differentiated material generation, routine assessment
- Humans handle: relationship building, motivational support, detecting frustration/anxiety, tier placement decisions, IEP compliance, instructional leadership during observations
- The "orchestrator" model: the teacher functions like a conductor, ensuring AI tools follow the pedagogical plan while maintaining authority over instructional decisions

Teacher Role During AI Tutoring

Role	Description
Orchestrator	Ensures AI tools follow the pedagogical plan; maintains instructional authority
Facilitator	Shifts from direct instruction to facilitating AI-supported learning opportunities
Interventionist	AI flags struggling students; teacher provides targeted human intervention
Quality Monitor	Reviews AI outputs, catches errors, ensures appropriateness
Relationship Builder	Focuses on motivation, confidence, and social-emotional support

Implementation Best Practices

1. Start with MTSS Infrastructure, Not AI

AI amplifies your existing processes. If your team structures, data review cycles, and Tier 1 instruction are weak, AI will not fix them. Ensure basic MTSS processes are functioning before adding technology.

2. Choose Evidence-Based, MTSS-Specific Platforms

Prioritize tools with ESSA evidence ratings. Check NCII tools charts for validated screening and progress monitoring tools. Prefer purpose-built MTSS platforms over generic AI for student-level decisions.

3. Ensure Adequate Dosage

Plan for at least 30 minutes per week of consistent usage. Build AI time into the master schedule rather than relying on optional or homework-based use. Monitor usage data weekly.

4. Invest in Ongoing Professional Development

Begin with addressing fear and comfort. Use a "train the trainer" model through instructional coaches. Establish PLCs for ongoing data interpretation and collaborative problem-solving. Budget for multi-year PD, not one-time workshops.

5. Maintain Human Decision-Making Authority

AI should inform, not replace, professional judgment in tier placement. Every AI recommendation should be reviewed by the MTSS team before implementation. This is both a best practice and a legal necessity for special education.

6. Audit for Bias Regularly

Check whether AI-flagged students match expected demographic profiles. Investigate any disparities in identification or recommendation patterns. Build fairness into AI selection criteria.

7. Ensure FERPA/COPPA Compliance

Verify that all AI platforms are FERPA-compliant and do not use student data to train external models. Never enter student PII into general-purpose AI tools. Document compliance as part of Tennessee's Public Chapter 550 AI policy requirements.

Funding Strategies Post-ESSER

- TISA direct funding: $500/student for rising 4th-grade literacy tutoring
- Title I: Best-known federal funding stream for tutoring in high-need schools
- Title III: Available for ELL-focused tutoring programs
- General fund: AI tools at $20-60/student/year are dramatically cheaper than human tutoring
- Braided funding: Combine Title I, general funds, state allocations, and nonprofit partnerships
- Niswonger Foundation: Potential local partner for NE Tennessee districts

The U.S. Department of Education's "Dear Colleague" letter explicitly authorized use of existing federal grants for AI integration in tutoring, with principles requiring initiatives be educator-led, ethical, accessible, and privacy-compliant.

8. Bottom Line

AI tutoring is real, it works, and it is entering K-12 classrooms at accelerating speed. The evidence base, while still young, is stronger than for most education technology interventions. Meta-analyses consistently show moderate positive effects (d = 0.43-0.76) for intelligent tutoring systems compared to conventional instruction, and emerging RCTs of generative AI tutors show promising results—particularly when combined with human tutors and strong pedagogical design. For a district like Greeneville City Schools, the question is no longer whether AI tutoring is worth considering, but how to implement it well.

The most important lesson from the research is that design matters more than technology. The same GPT-4 model produced learning gains exceeding one standard deviation at Harvard and a 17% decline in independent test performance in Turkey. The difference was entirely in pedagogical guardrails and implementation design. Districts that deploy AI tools without clear protocols, teacher training, and integration into existing MTSS processes are likely to see no benefit—or worse, see harm through cognitive dependency. Districts that invest in human-AI hybrid models, maintain teacher authority over instructional decisions, and ensure consistent dosage are likely to see meaningful gains, particularly for their lowest-performing students.

Tennessee is positioned relatively well for AI adoption. The state was among the first to require AI policies in schools (Public Chapter 550), TISA provides ongoing tutoring funding ($500/student for rising 4th-grade literacy), and the SCORE survey shows widespread educator engagement with AI tools. However, significant gaps remain: TDOE has not issued AI-specific guidance for RTI², no TDOE-approved AI platform list exists, and no Tennessee district under 8,600 students has publicly documented an AI tutoring implementation. Greeneville City Schools, with its strong TCAP trajectory and proximity to the Niswonger Foundation, has an opportunity to be a small-district leader in this space.

The honest assessment is that critical questions remain unanswered. No long-term (semester or year-long) RCTs of AI tutoring in K-12 have been published. The possibility of novelty effects fading, cognitive dependency developing, and equity gaps widening without deliberate intervention are all real risks that the current evidence cannot fully address. Khanmigo—the most widely deployed AI tutoring platform in K-12—has not yet completed its first RCT. The effect sizes that look impressive in controlled studies (d = 0.76-1.30) consistently shrink by one-third to one-half at district scale. A district leader should approach AI tutoring as a promising but unproven complement to proven high-dosage human tutoring, not as a replacement for it.

For Greeneville City Schools specifically, a prudent next step would be a small pilot at one or two grade levels in math or reading, using an evidence-rated platform (Amira for reading, Khan Academy or IXL for math), integrated into existing RTI² Tier II processes, with consistent 30+ minute weekly dosage, teacher training through PLCs, and pre/post assessment using TCAP-aligned measures. This approach would be low-cost (under $10,000 for a pilot), aligned with Tennessee's policy environment, and positioned to generate local data that informs scaling decisions.

Appendix A: Effect Size Summary Table

Note: Effect sizes below 0.20 are considered small, 0.20-0.50 medium, 0.50-0.80 large, and above 0.80 very large in education research. RCT designs provide causal evidence; quasi-experimental and observational designs show correlation only.

Study	Year	Design	N	Effect Size (d)	Comparison
VanLehn (human tutoring)	2011	Meta-analysis	Multiple	0.79	No tutoring
VanLehn (ITS)	2011	Meta-analysis	Multiple	0.76	No tutoring
Ma et al. (ITS overall)	2014	Meta-analysis	14,321	0.43	All comparisons
Kulik & Fletcher	2016	Meta-analysis	Multiple	0.66	Conventional
Carnegie Learning	2014	RCT	18,743	0.21 (Yr 2)	Conventional
ASSISTments (Maine)	2016	RCT	Large	0.18	No ITS
ASSISTments (NC)	2024	RCT	Large	0.10	No ITS
Khan Academy (RCT)	2024	RCT	~11,000	0.12-0.22	Control
Amira Learning	2024	Quasi-exp.	12 districts	0.40	Matched controls
Kestin/Harvard	2025	RCT	194	0.73-1.30	Active learning
Bastani (w/ guardrails)	2025	RCT	~1,000	+127% during	Control
Bastani (no guardrails)	2025	RCT	~1,000	-17% on exam	Control
Tutor CoPilot	2024	RCT	1,800	+4pp (+9pp)	Tutors w/o AI
LearnLM/Eedi	2025	RCT	165	+5.5pp	Human tutors
AI Meta-Analysis	2025	Meta-analysis	2,657	0.924	Traditional
GenAI Meta-Analysis	2025	Meta-analysis	68 studies	0.45	Various

Appendix B: AI Platform Comparison for RTI/MTSS

Platform	Function	ESSA Evidence	RTI Tier	Cost Model
Amira Learning (HMH)	AI reading tutor + assessment	Level II (Strong)	1, 2, 3	Per-student
Lexia Core5/PowerUp	Adaptive reading K-12	Strong (20 studies)	1, 2, 3	Per-student
Khan Academy / Khanmigo	AI tutor + practice	Observational	1, 2	Free / $35/student
IXL	Adaptive math/ELA K-12	ESSA Tier 2	1, 2, 3	Per-student
DreamBox Math	Adaptive math K-8	Strong ESSA	1, 2	Per-student
i-Ready (Curriculum Assoc.)	Adaptive assessment + instruction	Multiple ESSA	1, 2	Per-student
MAP Growth (NWEA)	Adaptive assessment	Validated screener	1 (screening)	Per-student
Classworks + Wittly	Assessment + AI intervention	Growth studies	1, 2, 3	Per-student
Branching Minds + Dottie	MTSS management + AI	Platform validation	All (mgmt)	Per-student
Panorama + Solara AI	MTSS/EWS platform	Platform validation	All (mgmt)	Per-student
Tutor CoPilot (Stanford)	AI-assisted human tutoring	RCT (gold standard)	2, 3	~$20/tutor/yr

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